

1-1-1: Introduction to Network Communications

At the end of this episode, I will be able to:

1. Identify and explain the purpose and importance of network components and communication.

Learner Objective: *Identify and explain the purpose and importance of network components and communication*

Description: In this episode, the learner will examine the basic components of modern networks .

- Purpose of Networks
 - o Facilitate data sharing and communication between devices.
 - o Enable access to shared resources and internet connectivity.
 - Transmission Media
 - o Copper Cables (e.g., Twisted Pair, Coaxial)
 - ♣ For wired connections.
 - o Fiber Optic Cables
 - ♣ For high-speed data transmission.
 - o Wireless Signals (e.g., Wi-Fi, Cellular Networks)
 - ♣ For mobile and remote access.
 - Endpoints
 - o Servers (e.g., Web, File, Email Servers)
 - ♣ To provide centralized services.
 - o Computers (e.g., Desktops, Laptops)
 - ♣ As primary user devices.
 - o Mobile Devices (e.g., Smartphones, Tablets)
 - ♣ For portable connectivity.
 - Redistribution Points
 - o Hubs
 - ♣ For basic signal retransmission in a network.
 - o Routers
 - ♣ For directing traffic at the network layer.
 - o Switches
 - ♣ For managing traffic at the data link layer.
 - Network Topology Considerations
 - o Bus, Star, Ring, Mesh, Tree, and Hybrid topologies
 - ♣ Affect layout and performance.
 - o Choice of topology
 - ♣ Impacts network scalability, redundancy, and maintenance.
 - Additional Components
 - o Network Interface Cards (NICs)
 - ♣ In devices for network connectivity.
 - o Access Points
 - ♣ For extending wireless network coverage.
 - o Firewalls
 - ♣ For securing networks against unauthorized access and threats.
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- Additional Resources:
 - o If applicable

